

Jermaine Tyrell Dennis

Full-Stack Software Engineer — 0→1 Product / Real-Time Systems

U.S. Citizen · San Francisco Bay Area · Available on-site

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Summary

Full-stack engineer who takes ideas from whiteboard → prototype → real users → deployment, solo. I designed and shipped a production platform end-to-end — Python async backend, React/TypeScript frontend, real-time sensing/inference (voice), data-collection + evaluation pipelines, experiments, and billing — with live users across 7 channels. I own features start to finish, make product calls under ambiguity, and ship fast without waiting for a spec. Strong in Python (async APIs, Pydantic-style typed models, pandas/numpy), modern TypeScript/React, real-time systems, and instrumentation/experiment loops.

What I've Built (Founder / Full-Stack Engineer, Hololoop Inc. — 2022–Present)

Yakimo — production cross-surface AI product with live users. Solo-architected and solo-built the entire stack.

0→1 product + user-facing features

- Took the product from empty repo to live users end-to-end: idea → prototype → in-the-wild user feedback → iteration → deployment, owning backend, frontend, infra, and release.
- Built a full-screen real-time chat/voice product experience (React 19 / Next.js 15 / TypeScript) with live typing/presence, animation, and an embeddable widget that drops into any external site via a postMessage bridge.

Data collection, experiments & instrumentation

- Built the internal **evaluation + data-collection platform** for the product: captures single- and dual-participant session transcripts, records interaction sessions, and runs structured evals used

to validate behavior and refine product direction — plus event/analytics instrumentation and a revenue dashboard I used to iterate on UX.

Real-time sensing/inference

- Built **3 swappable real-time voice pipelines** (OpenAI Realtime over WebRTC + server sideband, Cartesia cascaded STT→LLM→TTS over Twilio, Gemini Live) — real-time inference surfaced in a reliable, "feels-like-magic" product experience.
- Designed a retrieval system on **pgvector** with privacy/ACL scoping and lexical fallback; pub/sub event pipelines drive real-time UI (typing, presence, animation triggers).

Backend & data

- Python async backend: **Django (async) + Strawberry GraphQL**, typed schema/models, PostgreSQL + pgvector, Redis pub/sub, WebSocket subscriptions, connection-pool-aware background workers. (*Framework-agnostic on Python web stacks — comfortable in FastAPI/Pydantic; same async-API + typed-model patterns.*)
- Full **billing/transaction stack**: Stripe + PayPal, subscription tiers, credit enforcement across every channel, payouts.

Testing, CI/CD & infra

- Playwright **E2E test suites** (auth, websocket, chat flows) plus GraphQL smoke tests and regression coverage for fast iteration.
- **CI/CD on Render**, ASGI deployment, cron orchestration, Docker; management commands + recovery tooling for safe production operations.

Integrations & protocols

- Unified reply path delivering the same product across **web, phone/SMS (Twilio), Instagram, LINE, embed, and social feed**.
- **Bidirectional MCP** (agent as both tool-server and tool-client), plus a demoed paid agent-to-agent workflow driving a remote desktop (Blender) via MCP with an escrow hold/settle/refund flow.

Distribution I validated myself

- Shipped the **GDC 2026 Event Finder** (data → retrieval → embeddable widget): 800+ views, 10+ real user conversations, repeat requests to extend it.
- **~97/100 QR-scan conversion at SF Techweek** — validated a real-world acquisition mechanic live.

Selected Prior Experience

3DCloud / Programming.com — Frontend 3D Software Engineer (Contract) · Remote · 2025

Built web-based 3D product configurators used by Home Depot, Lowe's, and Howdens; debugged

enterprise-scale WebGL systems for reliability and performance. *React, TypeScript, C#, WebGL, real-time APIs.*

Square Enix (Japan) – Lighting/Compositing Artist & Python Pipeline Engineer · Tokyo · 2016–2018

Built Python automation tools and asset-management pipelines for *Final Fantasy* and *Kingdom Hearts*; repaired core render-farm workflows and shipped internal tools used daily by artist teams. Created the official *Final Fantasy VII Remake* key art (game covers / PlayStation Store).

Method Studios – Lighting Artist / Tools · 2020–2022 · *Thor: Love and Thunder* – Python automation for lighting/asset workflows on VFX-heavy sequences.

Self-Published AR/VR (Meta Start Developer) · 2021–2023 – Published VR apps (Oculus/Apple) in Unity/C#; real-time 3D environments, physics, and AI-driven characters. Systems/perf work on resource-constrained devices.

Technical Skills

Languages: Python, TypeScript/JavaScript, C# · (C/C++ familiar)

Backend: Python async APIs (Django async / FastAPI-style), typed models (Pydantic/GraphQL schema), PostgreSQL, pgvector, Redis, WebSockets, ASGI, REST & GraphQL · pandas/numpy

Frontend: React 19, Next.js 15 (App Router), TypeScript, Tailwind, Three.js / WebGL / Babylon.js

Real-time: WebRTC, WebSocket subscriptions (graphql-ws), pub/sub event pipelines, real-time inference (voice)

Data/experiments: event instrumentation, analytics dashboards, session/transcript capture, evaluation pipelines, A/B-style iteration loops

Testing & infra: Playwright E2E, smoke/regression tests, CI/CD, Docker, Render, AWS, cron orchestration

AI/LLM: OpenAI (Realtime/Chat/Embeddings), Gemini Live, Groq, RAG, MCP (bidirectional), prompt engineering

Education

BFA, Illustration – University of the Arts, Philadelphia, PA

Computer Graphics – Gnomon School of Visual Effects, Hollywood, CA